

SEM Wear Analysis Tool

A GUI-based tool for quantifying cutting tool wear from SEM images

1 What is it?

This tool analyses Scanning Electron Microscope (SEM) images of cutting tools to automatically measure how much of the tool surface is covered by different material layers. It identifies up to three distinct regions: **Coating**, **Adhesion layer**, and **Substrate**, and reports the area of each in square micrometres (μm^2).

It is designed to be used without any programming knowledge, through a simple graphical interface.

2 Why SEM images?

In an SEM image, different materials appear at different brightness levels. The outermost coating layer appears dark, the adhesion layer appears at a mid-grey level, and the substrate is the brightest region. This predictable relationship between material and brightness is what makes automated segmentation possible.

3 How does it work?

The analysis runs through four steps automatically once the user starts it.

1. **Tool isolation.** The tool surface is separated from the image background. Small noise regions are discarded and any holes inside the tool area are filled, so that only the actual tool surface is analysed.
2. **Noise cleaning.** Dark speckles that often appear inside bright material regions in SEM images are detected and removed. Each speckle is filled in using the surrounding pixel values so it does not affect the analysis.
3. **Finding the boundaries between materials.** The tool analyses the distribution of brightness values across the image. It uses a clustering algorithm (K-means) to automatically find the natural groupings of pixel intensities, each corresponding to one material. The boundaries between groups are then used to classify every pixel.

4. **Area calculation.** Each pixel is assigned to a material class based on its brightness. The number of pixels in each class is counted and converted to a physical area using a scale calibration provided by the user.

4 Using the interface

The interface is split into two phases.

Setup phase: before running the analysis, the user:

- Loads an SEM image.
- Draws a line over the scale bar visible in the image and enters its real length in micrometres. This calibrates the pixel to area conversion and is remembered when loading new images.
- Optionally crops the image to a specific region.
- Selects whether to segment 2 regions (Coating + Adhesion) or 3 regions (Coating + Adhesion + Substrate). This option is there because sometimes the tool surface might not be worn enough to have any substrate revealed.

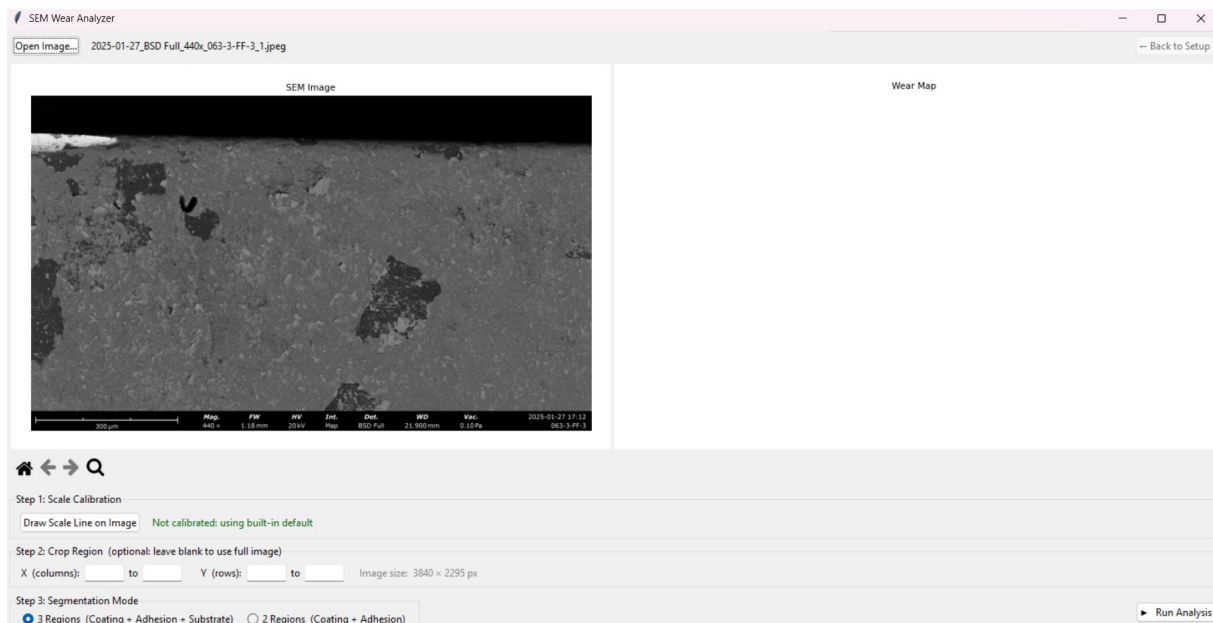


Figure 1: Setup phase: the user loads an image.

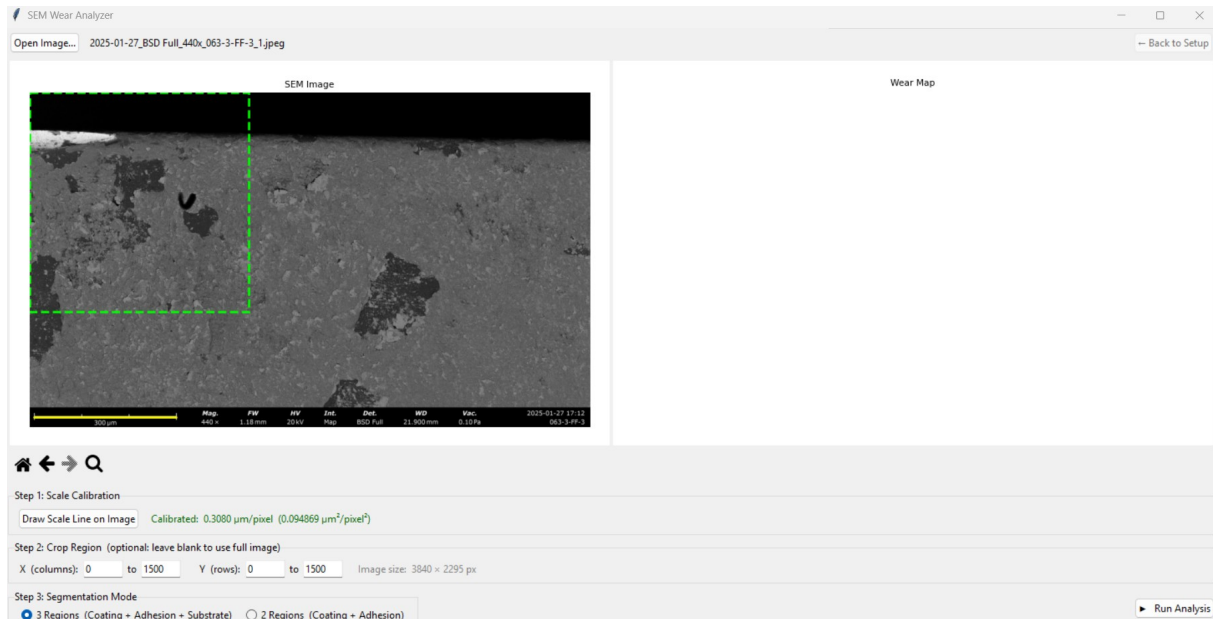


Figure 2: Setup phase: The user calibrates the scale, and configures the analysis.

Results phase: after the analysis completes:

- A colour-coded wear map is shown alongside the processed image (Blue = Coating, Red = Adhesion, Green = Substrate).
- The coverage percentage of each material is displayed.
- The user can fine-tune the boundary thresholds using sliders and see the wear map update instantly.
- All results can be saved to a folder: individual material masks, the wear map, and a text report with full area measurements.

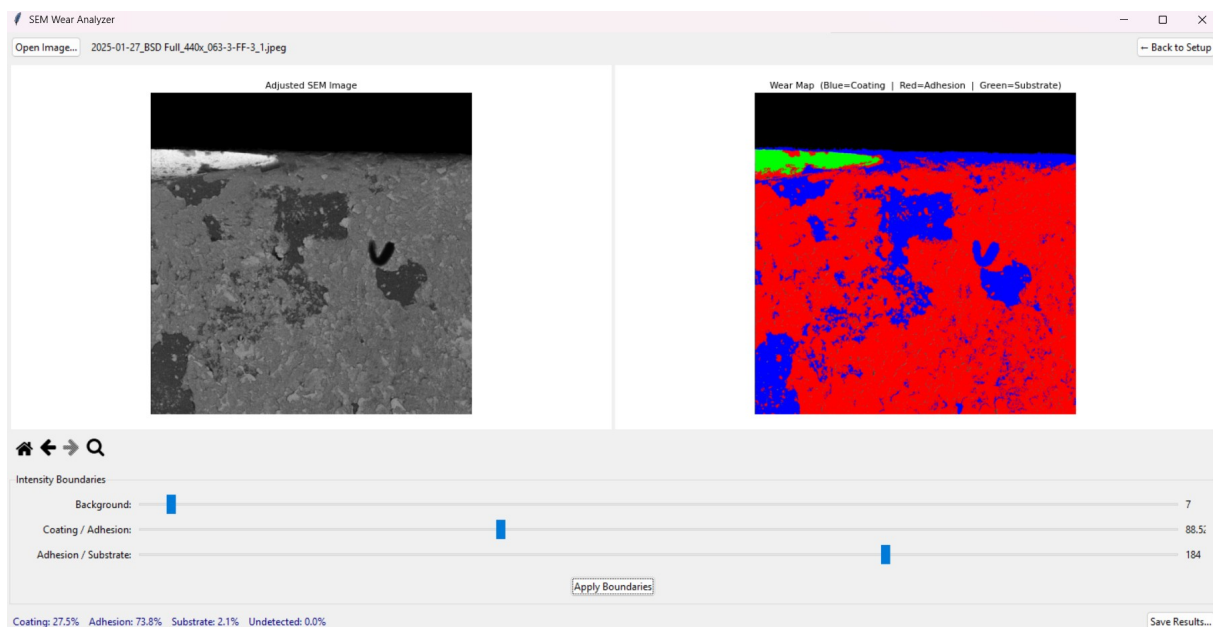


Figure 3: Results phase: colour-coded wear map and material area statistics.

5 Output

For each analysed image the tool produces:

- A colour wear map image.
- Individual binary mask images for each material region.
- A text report containing pixel counts, percentage coverage, and absolute areas in μm^2 for each material class.